Speaking script

Micro heat pipes and vapour chambers: liquid return in the wick, wick-thickness optimisation, vapour flow in a thin core. Debanshu Ghosh, 23CH3FP56. About 404 spoken words, 2:48 at 150 words a minute.

Slide	What it covers	Time	Running
1	Title	8 s	0:08
2	Overview: the capillary pump must beat every loss	28 s	0:36
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HOW TO USE THIS

- Say it, do not read the slide. The slide carries the picture and the formula; the script carries the sentence that connects them. The deck's speaker notes hold the same words, plus how each formula is derived.
- Each slide gets 24 to 34 seconds. If you are behind at the halfway mark (slide 4 should end at 1:34), drop the 37% sentence first, then the saturated-vapour sentence on slide 6.
- Rehearse once aloud with a timer. Silent reading is always faster than speaking.

NUMBERS TO HAVE IN YOUR HEAD

5× and 25×	pump pressure and liquid resistance, 10 µm against 50 µm powder
50, 38, 161 W	fine, coarse and biporous capillary limits at a 100 µm core (biporous is an upper bound)
37%	lower peak-to-mean condenser surface ΔT in Patankar's simulation, biporous condenser wick
50 µm, 150 µm, 101 W	best split of 250 µm for the 50 µm powder: wick layer, core, capillary limit
8×	vapour resistance when the core is halved ($1/h^3$)
1.1 K per kPa	saturation temperature change per vapour pressure drop, water at 60 °C
28× and 2×	chamber against copper sheet at 5 W, for h = 200 µm and h = 50 µm (the 50 µm value is a trend)

The calculated values are illustrative: water at 60 °C, a disc-shaped chamber with a 10 × 10 mm source and a 48 mm rim, 5 W. They are not measured device data. Say so if asked.

Slide 1 · Title — 8 s · ends 0:08

SAY

> Vapour chambers are the thin sheets that spread heat inside a phone. I will cover three transport questions about them.

POINT AT

- Stay on the title; the band shows vapour going right and liquid returning left.

Slide 2 · Overview: the capillary pump must beat every loss — 28 s · ends 0:36

SAY

> Heat evaporates water at the chip. Vapour flows through the core, condenses over the cooler surface, and the liquid returns through a porous wick. The only pump is the wick itself. So capillary pressure must beat the liquid loss, the vapour loss and gravity, and that sets a maximum heat load. My three topics are the terms of this one equation: the liquid return, the thickness split, and the vapour core.

POINT AT

- Trace the loop with your finger: chip, vapour arrow, heat out, liquid back through the wicks.
- Point at badges 1, 2, 3 in turn during the last sentence, then at the two formulas.

THE FORMULAS, IN ONE BREATH EACH

$$\Delta P_{cap} \geq \Delta P_l + \Delta P_v + \Delta P_g$$

Around a closed loop the pressure must balance. Only the meniscus raises it; liquid flow, vapour flow and gravity all lower it.

$$Q_{max} = \frac{\Delta P_{cap} - \rho_l \, g \, \Delta z}{a_l + a_v}$$

Both flow losses are proportional to the heat load Q (a is a flow resistance in Pa/W). Set the two sides equal and solve for Q.

IF ASKED

Why is there a gravity term?

If the chip is at the top of a vertical sheet the liquid must be lifted: ρ g Δz = 0.96 kPa for 100 mm. Horizontal, it is zero.

Why does orientation hardly matter inside the sheet?

The Bond number $\rho g h^2/\sigma$ is only 0.006 at 200 µm, so surface tension beats gravity at that scale.

Are the numbers from a real device?

No. They are for water at 60 °C and an illustrative phone-like disc: 10 × 10 mm source ($R1 = 5.64$ mm), rim at $R2 = 48$ mm, 5 W.

Slide 3 · Topic 1: finer pores pump harder but resist flow far more — 34 s · ends 1:10

SAY

> The meniscus in each pore is the pump, and Young–Laplace says its pressure goes as one over pore radius. But the liquid also has to flow through the wick, and Darcy's law gives that loss. Permeability goes as particle size squared, so one length sets both. Going from 50 to 10 micron powder gives 5 times the pump pressure but 25 times the liquid resistance. So the best wick is the coarsest one with enough margin.

POINT AT

- Left: the meniscus. Middle: coarse strip, then fine strip. Right: the three formulas top to bottom.
- Finish on the 5× and 25× tiles.

THE FORMULAS, IN ONE BREATH EACH

$$\Delta P_{cap} = \frac{2\,\sigma \cos\theta}{r_{eff}}$$

σ surface tension, θ contact angle, r_{eff} the effective pore radius. A spherical meniscus of radius $r_{eff}/\cos\theta$ has a pressure jump 2σ divided by that radius.

$$\Delta P_l = \frac{\mu_l\,\dot{m}\,\ln(R_2/R_1)}{2\pi\rho_l K\,t_l}$$

μ_l viscosity, $\dot{m} = Q/h_{fg}$ the mass flow, K permeability, t_l the total wick thickness (two layers). It is Darcy's law integrated radially from the rim R2 to the source R1.

$$K = \frac{d^2\varepsilon^3}{150(1-\varepsilon)^2}, \qquad r_{eff} \approx 0.21\,d$$

d particle diameter, ε porosity (0.5 here). With ε = 0.5 this is $K = d^2/300$. Pore radius is about 0.21 of the particle diameter.

IF ASKED

Why 0.21 d?

A common rule of thumb for sintered spheres; mesh and groove wicks have their own correlations. It is a stated assumption, not derived here.

What if the wick does not wet perfectly?

ΔP_{cap} carries cos θ, so θ = 60° halves the pump pressure with no change in geometry.

Is Darcy's law valid for such a thin wick?

It needs a layer many particles thick. A 50 µm layer of 50 µm powder is about one particle deep, so treat that case as an order-of-magnitude estimate.

Slide 4 · Topic 1: biporous wicks separate the pump from the flow path — 24 s · ends 1:34

SAY

A biporous wick escapes that trade-off. Fine pores inside clusters set the pump pressure, and coarse channels between clusters carry the flow. In my idealised estimate the limit rises from 50 or 38 watts to about 161, as an upper bound. In Patankar's simulation, a biporous condenser wick cut the peak-to-mean surface temperature difference by 37 percent.

POINT AT

- Illustration first (clusters and channels), then the three bars, then the formula, then the 37% tile.

THE FORMULAS, IN ONE BREATH EACH

$$Q_{max} = \frac{2\sigma / r_{fine}}{a_l(K_{coarse}) + a_v}$$

The pump comes from the fine pore radius, while the liquid resistance uses the coarse permeability. Each job gets its own length scale.

IF ASKED

Why only an upper bound?

It assumes the fine and coarse regions do not interfere, and it borrows K from 100 µm particles, which would not fit in a 50 µm layer. It shows the principle, not a buildable wick.

What exactly is the 37%?

A simulation result (Patankar, Weibel and Garimella 2016): the peak-to-mean temperature difference on the condenser surface, relative to a monolithic wick, because the biporous wick allows a thicker vapour core.

Slide 5 · Topic 2: in a fixed thickness every wick has a best split

32 s · ends 2:06

SAY

The inner thickness is fixed, here 250 microns, so every micron of wick comes out of the vapour core. Liquid resistance falls as one over t, but vapour resistance climbs as one over h cubed, so the total has a minimum. Setting the derivative to zero gives a clean result: h squared over t depends only on permeability and the fluid. For the 50 micron powder that means a 50 micron wick, a 150 micron core, and about 101 watts.

POINT AT

- Left: the three drawings, 57 W then 101 W then 27 W. Right: the curves; each dot is the best split for that wick.
- Point at the h²/t card last.

THE FORMULAS, IN ONE BREATH EACH

$$a_l + a_v = \frac{\ln(R_2/R_1)}{\pi h_{fg}} \left(\frac{\mu_l}{4\rho_l K t} + \frac{6\mu_v}{\rho_v h^3} \right)$$

The first term is the Darcy loss per watt with $t_l = 2t$ (it falls as 1/t). The second is the Poiseuille loss per watt (it rises as $1/h^3$). Together they are the total flow resistance, with $h = H - 2t$.

$$\frac{h^2}{t} = 12\sqrt{K\frac{\nu_v}{\nu_l}}$$

Differentiate with $dh/dt = -2$: the 3 from the cube and the 2 give 6, so $h^4/t^2 = 144\ K\ \nu_v/\nu_l$, and the square root gives the 12. $\nu = \mu/\rho$ is the kinematic viscosity.

IF ASKED

Does the optimum depend on the heat load?

No. Q, σ, θ and R2/R1 all cancel. Only the permeability and the fluid remain.

Is this from a paper?

No. It is my derivation from the same model, and I checked it against a direct numerical sweep for all three wicks (within the 2 µm grid).

Does it also minimise temperature?

No, it maximises the capillary limit only. At the optimum the vapour drop is 4.2 kPa, about 21% of P_{sat}, so read the watts as trends.

Slide 6 · Topic 3: halve the vapour gap and copper catches up

34 s · ends 2:40

SAY

Vapour spreads radially in a thin slit, which is Poiseuille flow, so the pressure drop goes as one over h cubed. Halve the gap and it is eight times worse. In saturated vapour that pressure drop is a temperature drop, about 1.1 kelvin per kilopascal. Against a copper sheet of the same thickness, at 5 watts, the chamber is about 28 times better at 200 microns, but only about 2 times better at 50. Thin cores erase the advantage.

POINT AT

- Left: radial picture, then the Poiseuille profile with the ×8 line. Then the two formulas.
- Right: grey bars are copper, dark bars are the chamber; finish on the tiles with the ratios.

THE FORMULAS, IN ONE BREATH EACH

$$\Delta P_v = \frac{6\mu_v \dot{m} \ln(R_2/R_1)}{\pi \rho_v h^3}$$

Plane Poiseuille flow in a slit gives dP/dr = −12 μ_v u/h²; the flow area is 2π r h, so u = ṁ/(ρ_v 2π r h). Integrate from R1 to R2. The h³ is one power from the velocity and two from the wall shear.

$$\Delta T_{sat} \approx \frac{T \Delta P}{\rho_v h_{fg}}$$

Clausius–Clapeyron with $v_{fg} \approx 1/\rho_v$. For water at 60 °C it is 1.08 K per kPa.

$$\Delta T_{Cu} = \frac{\dot{Q} \ln(R_2/R_1)}{2\pi k H}$$

The copper sheet of thickness H carries the same heat radially from the same source to the same rim. k = 400 W/m K. H is the whole chamber thickness (core, two wicks, two walls).

IF ASKED

What is the copper comparison exactly?

A solid copper sheet with the same total thickness H as the chamber, the same 5 W source and the same rim. At h = 200 µm: copper 9.3 K, chamber 0.33 K.

What is inside the chamber's ΔT?

Vapour drop + wick conduction + evaporating interface (accommodation coefficient assumed 0.1). At 200 µm that is 0.10 + 0.02 + 0.21 K. The interface term is the biggest for thick cores.

Why is the 50 µm bar marked with an asterisk?

Its vapour pressure drop (6.2 K equivalent) is above 20% of P_{sat}, so the linear Clausius–Clapeyron relation is not valid. Read it as a trend.

Is this a micro heat pipe or a vapour chamber?

A micro heat pipe is the one-dimensional version of the same physics. The numbers here are for the flat chamber geometry.

Slide 7 · Three results and references

8 s · ends 2:48

SAY

So: one length sets the wick, a simple formula sets the split, and h cubed sets the vapour limit. Thank you.

POINT AT

- Read the three badges quickly; leave the reference list on screen for questions.